QUESP and QUEST Revisited – Fast and Accurate Quantitative CEST Experiments

Moritz Zaiss,[1*†] Goran Angelovski,[2†] Eleni Demetriou,[3] Michael T. McMahon,[4,5] Xavier Golay,[3] and Klaus Scheffler[1,6]

Purpose: Chemical exchange saturation transfer (CEST) NMR or MRI experiments allow detection of low concentrated molecules with enhanced sensitivity via their proton exchange with the abundant water pool. Be it endogenous metabolites or exogenous contrast agents, an exact quantification of the actual exchange rate is required to design optimal pulse sequences and/or specific sensitive agents.
Methods: Refined analytical expressions allow deeper insight and improvement of accuracy for common quantification techniques. The accuracy of standard quantification methodologies, such as quantification of exchange rate using varying saturation power or varying saturation time, is improved especially for the case of nonequilibrium initial conditions and weak labeling conditions, meaning the saturation amplitude is smaller than the exchange rate ($\gamma B_1 < k$).
Results: The improved analytical 'quantification of exchange rate using varying saturation power/time' (QUESP/QUEST) equations allow for more accurate exchange rate determination, and provide clear insights on the general principles to execute the experiments and to perform numerical evaluation. The proposed methodology was evaluated on the large-shift regime of paramagnetic chemical-exchange-saturation-transfer agents using simulated data and data of the paramagnetic Eu(III) complex of DOTA-tetraglycineamide.
Conclusions: The refined formulas yield improved exchange rate estimation. General convergence intervals of the methods that would apply for smaller shift agents are also discussed.
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Key words: CEST; paraCEST; qCEST; QUESP; QUEST; Bloch-McConnell; quantification

INTRODUCTION

^{1}H chemical exchange saturation transfer (CEST) is an MRI technique capable of indirectly detecting millimolar concentrations of exchangeable protons with molar sensitivity via the water signal. Exchangeable protons of endogenous or exogenous compounds resonate at distinct frequencies relative to the water protons and can therefore be selectively saturated by radiofrequency irradiation. In addition, given the low solute concentration (μM–mM range) compared with water protons (approximately 111 M), prolonged saturation periods provide an amplification process for the small solute signal, which produces a reduction in water signal. This capability for amplified detection has allowed the design of new families of contrast agents, such as paramagnetic CEST agents, which enable the measurement of pH, temperature and buffer condition (1–7), or diamagnetic CEST agents such as the thymidine analogs (8), imidazoles, salicylates (9–11), and iodinated agents (12–14). These probes contain protons that resonate at multiple frequencies, thus enabling a multicolor imaging, similar to optical imaging agents (15–17).

Spectroscopy-based methods have been proposed for measuring the exchange rates of slowly exchanging systems (18,19); however, they are not suitable for fast exchange rate quantification due to severe line broadening. In contrast, it is possible to provide a quantification for experiments in this regime with CEST, and data can be interpreted by the Bloch-McConnell (BM) differential equations that incorporate parameters such as free precession, excitation and relaxation, as well as the exchange between different spin pools (20).

McMahon et al. were the first to develop an MRI-compatible method to measure exchange rate based on exploiting the influence of saturation time and saturation power on signal intensity (21). Using a simplified solution of the BM equations as described in the work of Zhou et al. (22,23), McMahon et al. calculated exchange rates by changing either the labeling efficiency (by the saturation pulse power) or the saturation time (21), and compared their results with the fitting of the BM differential equations in time, which forms a gold standard for quantification of CEST (24).

The existing analytical solutions that describe these experiments are well suited for calculation of exchange

[1]High Field Magnetic Resonance, Max Planck Institute for Biological Cybernetics, Tübingen, Germany.
[2]MR Neuroimaging Agents, Max Planck Institute for Biological Cybernetics, Tübingen, Germany.
[3]Brain Repair & Rehabilitation, Institute of Neurology, University College London, United Kingdom.
[4]Division of MR Research, The Russell H. Morgan Department of Radiology and Radiological Science, Johns Hopkins University, Baltimore, Maryland, USA.
[5]F.M. Kirby Research Center for Functional Brain Imaging, Kennedy Krieger Institute, Baltimore, Maryland, USA.
[6]Department of Biomedical Magnetic Resonance, Eberhard Karl's University of Tübingen and University Hospital, Tübingen, Germany.

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†These authors contributed equally to this work.
*Correspondence to: Moritz Zaiss, Ph.D., High Field Magnetic Resonance, Max Planck Institute for Biological Cybernetics, Spemannstrasse 41, Tübingen, Germany. E-mail: moritz.zaiss@tuebingen.mpg.de.

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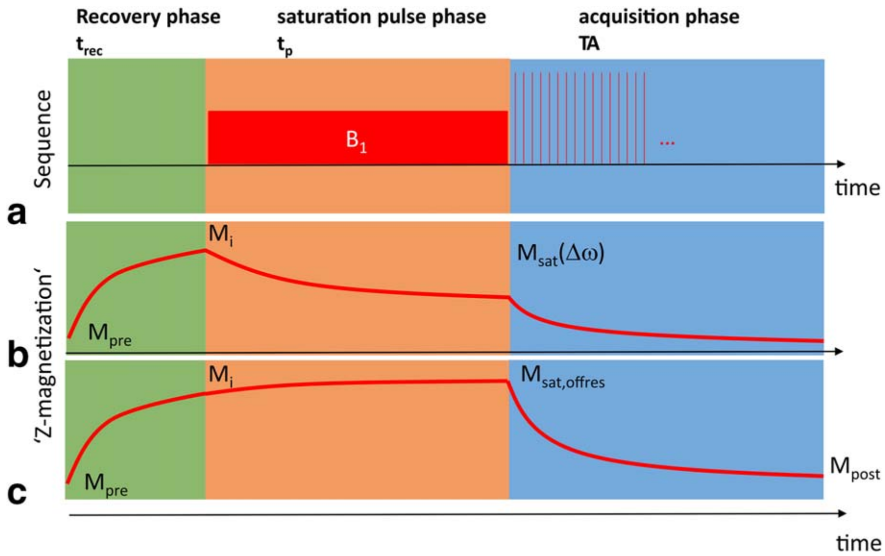

FIG. 1. **a**: The sequence diagram of a typical CEST experiment consists of three modules: recovery module of duration t_{rec}, saturation module of duration t_p, and acquisition module of duration TA. **b**: Magnetization course during continuous-wave irradiation of amplitude B_1: If $t_{rec} << 5 \times T_{1a}$, the Z-magnetization recovers to $M_i \neq M_0$; if $t_p << 5 \times T_{1a}$, the M_{sat} can depend on M_i. **c**: Magnetization course for far off-resonant irradiation: The course during t_p is also approximately governed by T_1 and therefore approaches M_0. If $t_{rec} + t_p << 5 \times T_1$, this value can still be $M_{offres} < M_0$. Given that $Z(\Delta\omega) = M_{sat}(\Delta\omega)/M_0$, the Z-value notation of the z-magnetization will be used in the rest of the manuscript.

rates, as long as a strong labeling is achieved. "Strong labeling" means in this context that the saturation amplitude is larger than the exchange rate ($\gamma B_1 > k_b$); thus, the magnetization of the CEST pool is close to 0 during irradiation, and the CEST effect is maximized as well as the labeling efficiency α, which approaches 1.

In this work we show that weak labeling ($\gamma B_1 < k_b$, $\alpha < 0.5$), as well as nonequilibrium initial magnetization conditions, may result in inaccurate quantification of exchange rates if the existing analytical solutions are used. Therefore, we focused on this issue and derived analytical quantification of exchange rate using varying saturation power (QUESP) and quantification of exchange rate using varying saturation time (QUEST) formulas based on the equivalence of the spin-lock theory to CEST experiments (25,26). The derived equations were applied to a system with large chemical shifts, in which direct water saturation is negligible. We show that these formulas agree with the original QUESP/QUEST equations for strong labeling; furthermore, they extend the convergence interval for weak labeling. In addition to QUESP, we apply the similar Ω-plot method, which reformulates QUESP to a linear regression problem (27) using the same data. For the experimental validation of our formulas, we used paramagnetic CEST (paraCEST) agents (28) for the following reasons. First, these agents induce large shifts for exchanging protons, simplifying the theory (direct saturation does not have to be taken into account). Second, the knowledge of the exact exchange rate is very important for the design of novel and potentially useful classes of paraCEST molecules; therefore, they are the ideal test molecules for this application (29,30).

In this work, we show that data interpretation can be incorrect for this simple case, thus indicating the importance of extending the quantification methodology at this level. Moving to more complex cases and CEST effects closer to bulk water frequency is even more challenging, and these aspects will be analyzed in a forthcoming study. Moreover, the aim of this paper is to provide researchers interested in quantitative CEST experiments with a source code that is freely available for deducing exchanging rates and for reproducing the results of this article.

THEORY

The results of CEST experiments are usually presented in so-called Z-spectra. To obtain a Z-spectrum, the amplitude of the saturated water magnetization ($M_{sat}(\Delta\omega)$), normalized as a proportion of the unsaturated water magnetization (M_0), is plotted as a function of the saturation pulse frequency ω. As the acquired signal S_{sat} or S_0 is proportional to the magnetization, the normalized Z-spectrum is described by the value $Z(\Delta\omega)$ given as Equation [1] (31):

$$Z(\Delta\omega) = S_{sat}(\Delta\omega)/S_0 = M_{sat}(\Delta\omega)/M_0 \qquad [1]$$

A typical pulse-sequence diagram that is being used for quantitative CEST experiments and the principle behavior in time of the Z-magnetization is presented in Figures 1a and 1b, respectively. The depicted diagram shows the preparation at one frequency offset, with subsequent readout of the water signal either by a free induction decay (FID) or a fast imaging readout. Assuming a dummy scan, the magnetization M_{pre} recovers during t_{rec} time with relaxation defined by the water-relaxation time T_{1a} to the value M_i, which is the initial magnetization right before the saturation module (i.e., $M_i \neq M_0$). During the saturation pulse of duration t_p, the magnetization follows an $R_{1\rho}$ decay, reaching the Z-magnetization with the value M_{sat} right after irradiation. For long enough saturation conditions (e.g., $t_p >> T_{1a}$), M_{sat} approaches the saturation steady state. During the acquisition module, the Z-magnetization decreases to M_{post} (e.g., a short 90° pulse would set it to zero, or a gradient-echo readout would approach an imaging steady state). Finally, after the readout the magnetization is M_{post}. The CEST experiment for a single offset is now complete. However, to acquire a full Z-spectrum, the saturation frequency is now changed and the sequence is played out again with the condition $M_{pre} = M_{post}$ (assuming that M_{post} does not depend on M_{sat}).

In many cases, instead of using M_0, the data are often normalized using the signal acquired after far off-resonance irradiation (M_{offres}). This can lead to problems if the sum of saturation time and recovery time is shorter than T_{1a} (Fig. 1c). Then $M_{offres} < M_0$ and all normalized effects are artificially increased. In such case, as shown in detail in the Supporting Figures S1–S3, the quantification equations are not directly applicable. However, even if an additional M_0 signal is measured after long relaxation in a separate scan, the timing of the CEST sequence can still lead to an initial magnetization before saturation $M_i < M_0$, which can influence the saturated signal. Only after a long recovery time, before each saturation ($t_{rec} > 5 \cdot T_{1a}$) all previous magnetization history is lost regardless of the sequence timing (so $M_i = M_0 = M_{offres}$). Generally, both M_0 and M_i must be known to be able to interpret quantitative CEST experiments for arbitrary initial and saturation conditions. If t_p is short compared with T_{1a}, the measured value will depend on the initial value M_i. If t_p is long compared with T_{1a}, the magnetization approaches the saturation steady state.

To evaluate the CEST signal, two Z-values are commonly used: the label scan, which is the normalized Z-magnetization after saturation at the CEST resonance ($Z_{lab} = Z(\Delta\omega = +\delta\omega_b)$), and the reference scan, which represents the saturation without CEST effect that can be estimated from the baseline or, as here, from the opposite frequency ($Z_{ref} = Z(\Delta\omega = -\delta\omega_b)$).

The CEST effect can then be defined by the magnetization transfer ratio asymmetry (MTR_{asym}, Eq. [2]) or the inverse asymmetry of the normalized Z-magnetization (MTR_{Rex}, Eq. [3]):

$$MTR_{asym} = Z_{ref}(B_1) - Z_{lab}(B_1) = \frac{M_{sat}(-\delta\omega_b)}{M_0} - \frac{M_{sat}(+\delta\omega_b)}{M_0} \quad [2]$$

$$MTR_{\mathrm{Rex}} = \frac{1}{Z_{lab}(B_1)} - \frac{1}{Z_{ref}(B_1)} = \frac{M_0}{M_{sat}(+\delta\omega_b)} - \frac{M_0}{M_{sat}(-\delta\omega_b)} \quad [3]$$

Detailed theoretical descriptions and derivations of the Z-values and MTR_{asym} are given in the Appendix. In this section, we provide only the final formulas that we used for data evaluation. In addition, the initial Z-magnetization before saturation is written as $Z_i = M_i/M_0$ in the following sections.

Analytical Solution for Equilibrium Magnetization M_0 as the Initial Magnetization M_i

We consider a two-pool system of the water pool (Pool a) with thermal magnetization M_{0a}, and the CEST pool (Pool b) with thermal magnetization M_{0b}, and the relative fraction $f_b = M_{0b}/M_{0a}$. We assume that the initial magnetization at thermal equilibrium is $M_i = M_0$; thus, $Z_i = 1$. It can be shown that then and only then the CEST effect can be described quantitatively by Equation [4], or in other words, only then Equation [A10] simplifies to Equation [4]:

$$\begin{aligned} MTR_{asym}(\alpha(B_1), t_p) &= \frac{R_{ex}^{lab}}{R_{1a} + R_{ex}^{lab}}\left(1 - e^{-(R_1 + R_{ex}^{lab})t_p}\right) \\ &= \frac{f_b k_b \cdot \alpha}{R_{1a} + f_b k_b \cdot \alpha}\left(1 - e^{-(R_{1a} + f_b k_b \cdot \alpha)t_p}\right) \end{aligned} \quad [4]$$

For the so-called labeling efficiency α, different limits are published (31); we assume here large shifts between water and the CEST pool, then α reads as in Equation [5], where k_b is the exchange rate, R_{2b} is the transversal relaxation rate of the CEST pool, and the $\omega_1 = \gamma B_1$ is the radiofrequency saturation amplitude.

$$\alpha(B_1) = \frac{\omega_1^2}{\omega_1^2 + k_b(k_b + R_{2b})} \quad [5]$$

In the original QUEST/QUESP paper, the authors introduced MTR_{asym}, given by Equation [6] as

$$MTR_{asym} = \frac{f_b k_b \cdot \alpha}{R_{1a} + f_b k_b}\left(1 - e^{-(R_{1a} + f_b k_b)t_p}\right) \quad [6]$$

When comparing Equation [6] to Equation [4], the two additional α factors appear that scale the product of the fractional concentration and the exchange rate $f_b k_b$.

In this manuscript, we show that by using Equation [4] and including the α terms, improved estimates of the exchange rates can be calculated. Note also that Equation [6] becomes equal to Equation [4] for strong labeling: when $\alpha = \frac{\omega_1^2}{\omega_1^2 + k_b^2} \approx 1$, and when $R_{1a} >> k_b f_b$.

Using Equation [4] or [6], two experiments can be designed to quantify exchange rates from the water. A QUESP experiment can then be understood as acquisition of $MTR_{asym}(B_1)$ for varying B_1 at a fixed saturation duration t_p, whereas a QUEST experiment can be understood as $MTR_{asym}(t_p)$ for varying t_p at a fixed saturation amplitude B_1.

Equation [6] can be used for arbitrary saturation times t_p; however, for the sake of completeness we provide the original and revised QUESP Equations [7] and [8], respectively, in steady state (i.e., $t_p \to \infty$) as

$$MTR_{asym} = \frac{f_b k_b \cdot \frac{\omega_1^2}{\omega_1^2 + k_b^2}}{R_{1a} + f_b k_b} \quad [7]$$

$$MTR_{asym} = \frac{f_b k_b \cdot \frac{\omega_1^2}{\omega_1^2 + k_b^2}}{R_{1a} + f_b k_b \cdot \frac{\omega_1^2}{\omega_1^2 + k_b^2}} \quad [8]$$

This theory can be used in two ways to yield correct estimates of exchange rates. First, if the recovery time t_{rec}

(i.e., delay time before the saturation) (Fig. 1) is long enough so that the initial magnetization is fully relaxed (i.e., $Z_i = M_i/M_0 = 1$), then arbitrary saturation times t_p can be used and fitted by Equation [4]. Second, if the saturation time t_p is long enough ($> 3 \times T_{1a}$) so that the saturation steady state is reached and therefore is independent of Z_i, then exchange-rate quantification via QUESP experiments is possible using Equations [7] and [8].

Omega Plot Methods for Steady-State QUESP

In saturation steady state, using the inverse asymmetry MTR_{Rex}, exchange rates can also be calculated using Equation [9], which eliminates spillover and semisolid magnetization transfer, and relates MTR_{Rex} to k_b, ω_1, f_b, and R_{1a} as follows:

$$MTR_{\mathrm{Rex}} = \frac{1}{Z_{lab}(B_1)} - \frac{1}{Z_{ref}(B_1)} = \frac{1}{R_{1a}} f_b k_b \cdot \frac{\omega_1^2}{\omega_1^2 + k_b^2} \qquad [9]$$

In addition, using $1/MTR_{Rex}$, the exchange rate can be obtained from the x-intercept of a plot of steady-state CEST intensity as a function of $1/\omega_1^2$. The derived Equation [10] was given by Meissner et al. (33) as follows:

$$y\left(\frac{1}{\omega_1^2}\right) = \frac{1}{\frac{1}{Z_{lab}} - \frac{1}{Z_{ref}}} = \frac{\mathrm{R_{1a}}}{f_b k_b} + \frac{\mathrm{R_{1a}} k_b}{f_b} \cdot \frac{1}{\omega_1^2} \qquad [10]$$

Equation [10] is referred as the Ω-plot method and was originally introduced by Dixon et al. (27). The original Ω-plot formula is given by Equation [11], which is a special case of Equation [10] for $Z_{ref} = 1$:

$$y\left(\frac{1}{\omega_1^2}\right) = \frac{Z_{lab}}{1 - Z_{lab}} = \frac{1}{\frac{1}{Z_{lab}} - 1} = \frac{\mathrm{R_{1a}}}{f_b k_b} + \frac{\mathrm{R_{1a}} k_b}{f_b} \cdot \frac{1}{\omega_1^2} \qquad [11]$$

Analytical Solution for Arbitrary Initial Magnetization M_i

The theory as described previously can be extended to account for nonthermal equilibrium initial magnetization Z_i by using Equation [A10]; with R_{ex} (Eq. [A7]) inserted, it results in Equation [12], which shows an explicit dependency on the initial magnetization before the saturation module $Z_i = M_i/M_0$ (Fig. 1b):

$$\begin{aligned} MTR_{asym} &= Z_{ref}(t_p, B_1) - Z_{lab}(t_p, B_1) \\ &= \frac{f_b k_b \cdot \alpha}{R_{1a} + f_b k_b \cdot \alpha} + (Z_i - 1) e^{-R_{1a} t_p} \\ &\quad - \left(Z_i - \frac{R_{1a}}{R_{1a} + f_b k_b \cdot \alpha}\right) e^{-(R_{1a} + f_b k_b \cdot \alpha) t_p} \end{aligned} \qquad [12]$$

Therefore, for fast and accurate quantitative experiments, we suggest the following:

1. Measure M_0 for a recovery time t_{rec} equal to $5 \times T_{1a}$, and then use this M_0 for normalization of all Z-spectra. Alternatively, $M_{far\text{-}offres}$ for very long saturation time ($5 \times T_{1a}$) can be used as M_0 estimation;
2. Measure the initial magnetization right before saturation $Z_i = M_i/M_0$. This can be achieved by running the sequence with the same recovery timing, but removing the saturation block;
3. Measure the Z-spectra with decreased recovery and saturation times as long as the signal-to-noise ratio is sufficient;
4. Measure the T_1 of the sample; and
5. Fit the data by using the full BM equations, or Equation [12] and employ the measured R_{1a}, M_i, and M_0.
6. As described in the Supporting information, the normalization of the data and the fit must be the same.

METHODS

Simulation and Full BM Fitting

Numerical simulations and experimental data were processed using custom-written scripts in MATLAB version 8.2.0.701 (The MathWorks, Natick, MA) following previously reported procedures (24). To evaluate the revised formulas (Eqs. [4], [6], and [12]), the CEST data were simulated using a rectangular pulse shape with pulse duration t_p and amplitude B_1. The CEST effect was evaluated using an exchanging system of $k_b = 9000$ and 1000 Hz, respectively. The relative proton concentration f_b corresponds to a typical concentration of 15 mM: $f_b = 15\,\mathrm{mM}/111\,\mathrm{M} = 0.000135$. In addition, the water-pool relaxation is chosen similar to typical values of phantoms with $R_{1a} = 0.3$ Hz and $R_{2a} = 0.5$ Hz, and relaxation parameters of the CEST pool were set fixed to $R_{1b} = 1$ Hz, $R_{2b} = 50$ Hz. The static magnetic field was set to $B_0 = 7.0$ T. To sample the chemical shift of a CEST agent at 50 ppm, Z-spectra were sampled between −80 and 80 ppm. Rician noise of 0.1% of M_0 was added to the simulated Z-values. This was repeated for different saturation powers B_1 between 10 and 35 μT, and saturation times t_p between 0.5 and 10 s.

For fitting of experimental data, the two-pool system was extended to a three-pool model; the third amide pool resonating upfield from the water was added to the fitting model according to Dixon et al. (27) and (34); and the additional amide pool was initialized by the parameters $\delta\omega_{amd} = -6.0$ ppm, $f_{amd} = 0.5‰$, $k_{amd} = 50$ Hz. The paraCEST pool quantification was not influenced by $R_{2amd} = 50$ Hz.

Analytical solutions of the BM equations were fitted in MATLAB using the optimization function `lsqcurvefit`. All simulation and evaluation files can be found and downloaded from the following websites: cest-sources.org or https://github.com/cest-sources/BM_sim_fit.

Preparation of ParaCEST Contrast Agent

The experimental results reported in this work were obtained using EuDOTAM-Gly, a complex of Eu(III) with the tetraglycineamide derivative of DOTA. This complex was prepared according to the previously reported procedure (2). The agent was dissolved in H_2O:D_2O (9:1, v/v) at a concentration of 10 mM and pH = 7.4.

The CEST Experiments

All experiments were performed on a 7T (300.17 MHz) Bruker Avance III NMR spectrometer (Bruker, Ettlingen, Germany) using a rectangular saturation pulse followed by a 90° FID readout. The recovery time t_{rec} after readout was 3-6 s. The saturation transfer experiments were carried out at a temperature range of 10 to 50°C by irradiating the sample at increments of 1 ppm with frequency range ±80 ppm. Spectra were measured by recording the bulk water signal intensity as a function of the presaturation frequency. Saturation offsets are reported relative to the signal of bulk water. The used temperatures were corrected by measuring the frequency difference of neat ethylene glycol at each temperature reported (35).

For each temperature, data were collected by varying the saturation power while the saturation time remained constant (10 s). The saturation field strengths used were 10, 15, 20, 25, 30, and 35 μT. Longitudinal relaxation times were obtained in an independent experiment using the standard inversion recovery with 1% gradient to eliminate the radiation damping effect. Steady-state CEST experiments were performed using a 5-mm NMR tube; however, for CEST experiments with short saturation ($t_{rec} = 1$ s, $t_p = 3$ s) the volume was reduced by a factor of 0.1 to avoid radiation damping influences by using a smaller 2-mm NMR tube (1.6 mm inner diameter) filled with the sample, inserted into the 5-mm NMR tube filled with D_2O.

Data Evaluation

For data evaluation, the CEST effect was calculated using the asymmetry analysis MTR_{asym} or the inverse asymmetry of the normalized Z-magnetization MTR_{Rex}. Because the label and reference values are defined as $Z_{lab} = Z(\Delta\omega = +\delta\omega_b)$ and $Z_{ref} = Z(\Delta\omega = -\delta\omega_b)$, the MTR_{asym} and MTR_{Rex} are calculated using Equations [2] and [3].

Validation of QUEST/QUESP

For arbitrary saturation but thermal initial conditions ($Z_i = 1$), the original QUESP/QUEST equation given by Equation [6] was compared with the revised version (Eq. [4]).

For arbitrary initial conditions ($Z_i \neq 1$), the results obtained using Equation [12] were compared with the outcome of Equations [4] and [6], respectively. In all simulations and equations, negligible direct saturation of the water protons was assumed, and R_{2b} was assumed to be negligibly small compared with k_b and was set to 0.

Exchange Rate as a Function of Temperature

The Arrhenius Equation [13] allows the determination of the exchange rate constant of a first-order exchange process as a function of temperature T(36), in which k_c (298.15 K) is the collision frequency factor at 298.15 K, $E = E_A + \Delta H_R^0$ with the activation energy E_A, ΔH_R^0 is the standard reaction enthalpy for the self-dissociation of water (55.84 $\frac{kJ}{mol}$) (37), and R is the ideal gas constant (8.314 $\frac{J}{mol\ K}$).

$$k_b(T) = k_c(298.15\ K) \cdot \frac{mol}{l} \cdot 10^{\frac{E}{R \cdot \ln 10}\left(\frac{1}{298.15\ K} - \frac{1}{T}\right)} \quad [13]$$

RESULTS

The principle methods for quantification of exchange rates by varying saturation power (QUESP) or varying saturation time (QUEST) are shown in the Figures 2a and 2b, or 2c and 2d, respectively. In addition, the CEST Z-spectra at constant time and power with varying the exchange rate are shown in Figures 2e and 2f. The comparison in Figures 2b, 2d, and 2f of the analytical descriptions of MTR_{asym} (lines) with BM simulation (circles) verifies that the revised equation (Eq. [4]) improves the original Equation [6] (dashed lines in Figs. 2b, 2d, and 2f), especially in the case of weak labeling ($\gamma B_1 < k$).

The QUESP Approach and Ω-Plot Method in Steady State

Simulated Z-spectra data of two different k_b values of 1000 and 9000 Hz, using saturation powers of $B_1 = 5$ to 35 μT in steps of 5 μT and saturation time $t_p = 100$ s (steady state) were used to evaluate the simplified QUESP solutions and the Ω-plot method (Fig. 3). The obtained results indicate that the original QUESP formula (Eq. [6]) leads to overestimated f_b and underestimated k_b values (blue lines, Figs. 3a and 3b) when compared with the revised QUESP formula (Eq. [4] red line, Figs. 3a and 3b), which shows an improved estimation, closer to the actual concentration ($f_b = 1.35 \times 10^{-4}$) for both exchange rates. Consequently, the inclusion of the α factor in Equation [4] which is much smaller than 1 (i.e., $\alpha = 0.35$), is the key parameter that enables accurate estimates of exchange rates.

For steady-state conditions, the QUESP approach using inverse asymmetry (Figs. 3c and 3d), as well as the Ω-plot method (Figs. 3e and 3f), yielded the best estimation for $k_b = 9000$ Hz and satisfactory results for $k_b = 1000$ Hz. One has to keep in mind that the statistical errors for the Ω-plot method are different for each point in the plot and should be used as weights for the linear fit. Otherwise, the highest data point (which results from the lowest power) has too much influence on the least-squares fit; therefore, the outcome can be biased. In this work, we assumed constant noise and used MTR (or 1/y) as weights for the least-squares fit.

Non-Steady-State, Fully Relaxed Initial Condition for QUEST

Unlike QUESP, the QUEST analysis using a single offset does not allow for a full quantification of f_b, k_b and R_{2b}, as no separation of k_b and f_b in R_{ex} (see Eqs. [A4] and [A7]) is possible by QUEST. Consequently, the concentration f_b must be provided to the single offset QUEST fit to execute the exchange rate estimations.

Simplified QUEST Solution

Similar to the QUESP tests, we performed QUEST evaluations for two different k_b values (1000 and 9000 Hz, respectively) for saturation power of $B_1 = 10$ μT, and the

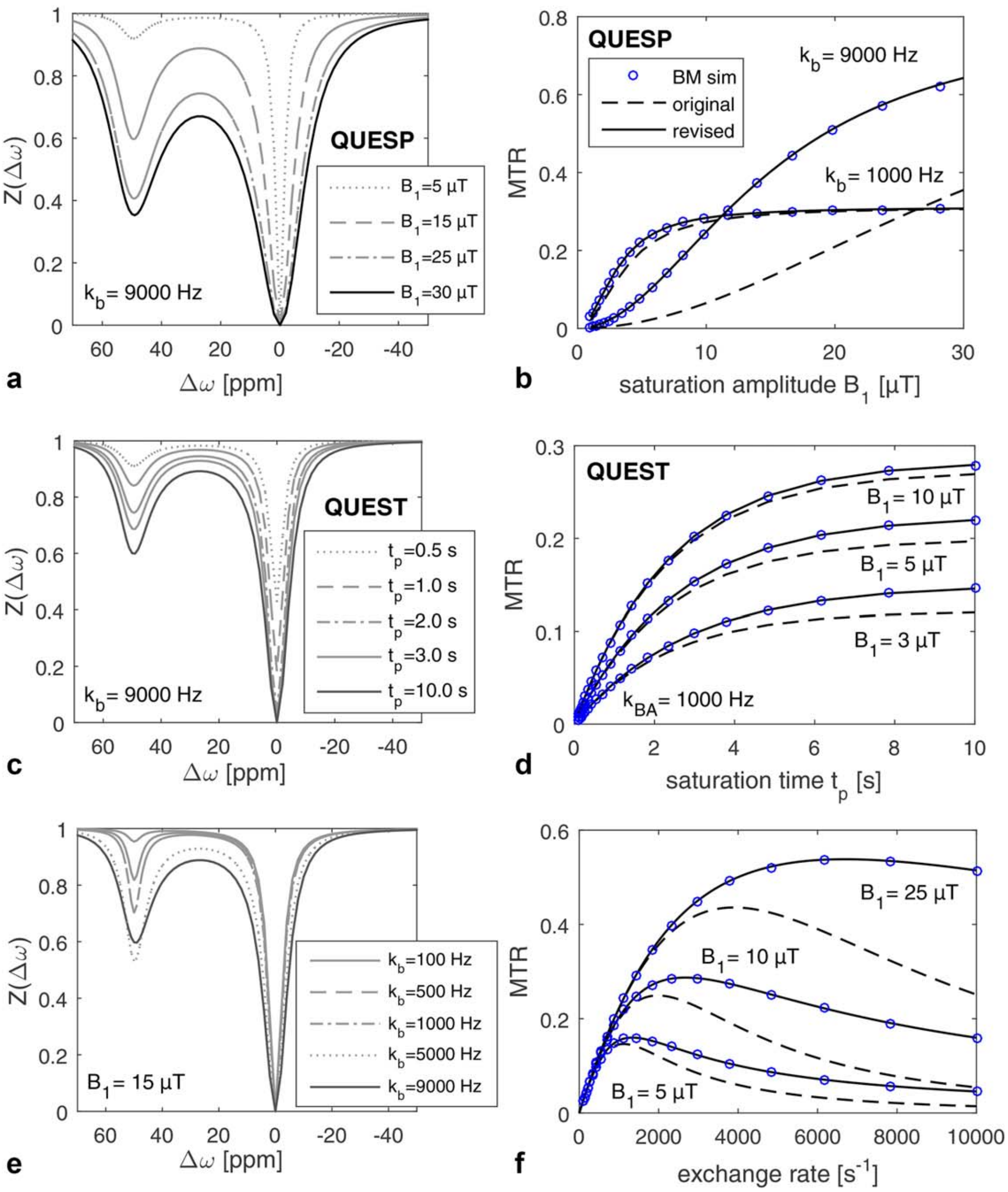

FIG. 2. Simulated Z-spectra obtained by varying saturation power B_1 (**a**), saturation time (**c**), and varying exchange rate (**e**). **b, d, f**: Comparison of BM simulation (circles) with different analytical QUESP/QUEST formulas (dashed line, Eq. [6]; solid line, Eq. [4]) for varying saturation power (**b**), saturation time (**d**), and varying exchange rate k (**f**) for $t_p = 3$ s. As the labeling efficiency term α was added in the revised formula (Eq. [4]), the two equations match when $k_b << \gamma B_1$, and thus when full labeling is achieved. In the case of weak labeling ($\alpha < 0.5$), the revised equation yields a much better match with the BM simulation. The curves in (**b**) for 1000 Hz also match both Equations [4] and [6] for $\alpha << 1$; this can be explained by the fact that $k_b \cdot f_b = 0.135 < R_{1a} = 0.33$ (hence the minor influence of the α terms in this case).

saturation times 0.5, 0.75, 1, 2, 3, 4, 5, 7.5, and 10 s. The relative concentration of $f_b = 1.35 \times 10^{-4}$ has been already provided and fixed (Fig. 4). Both solutions using the original and revised QUEST formulas (Eqs. [6] and [4], respectively) provide similar results for the lower exchange rate (Fig. 4a). However, the original QUEST formula (blue line, Eq. [6]) does not fit the data for given f_b, and underestimates k_b with 3900 Hz for the higher exchange rate of 9000 Hz (Fig. 4b). In contrast, the revised QUEST formula (Eq. [4]) yields an improved estimation with an exchange rate of 9139 ± 87 Hz.

Arbitrary Initial and Saturation Conditions With QUESP and QUEST

Until now, the important conditions of a fully relaxed initial magnetization (and thus long recovery time) or a steady-state saturation (and thus long saturation time)

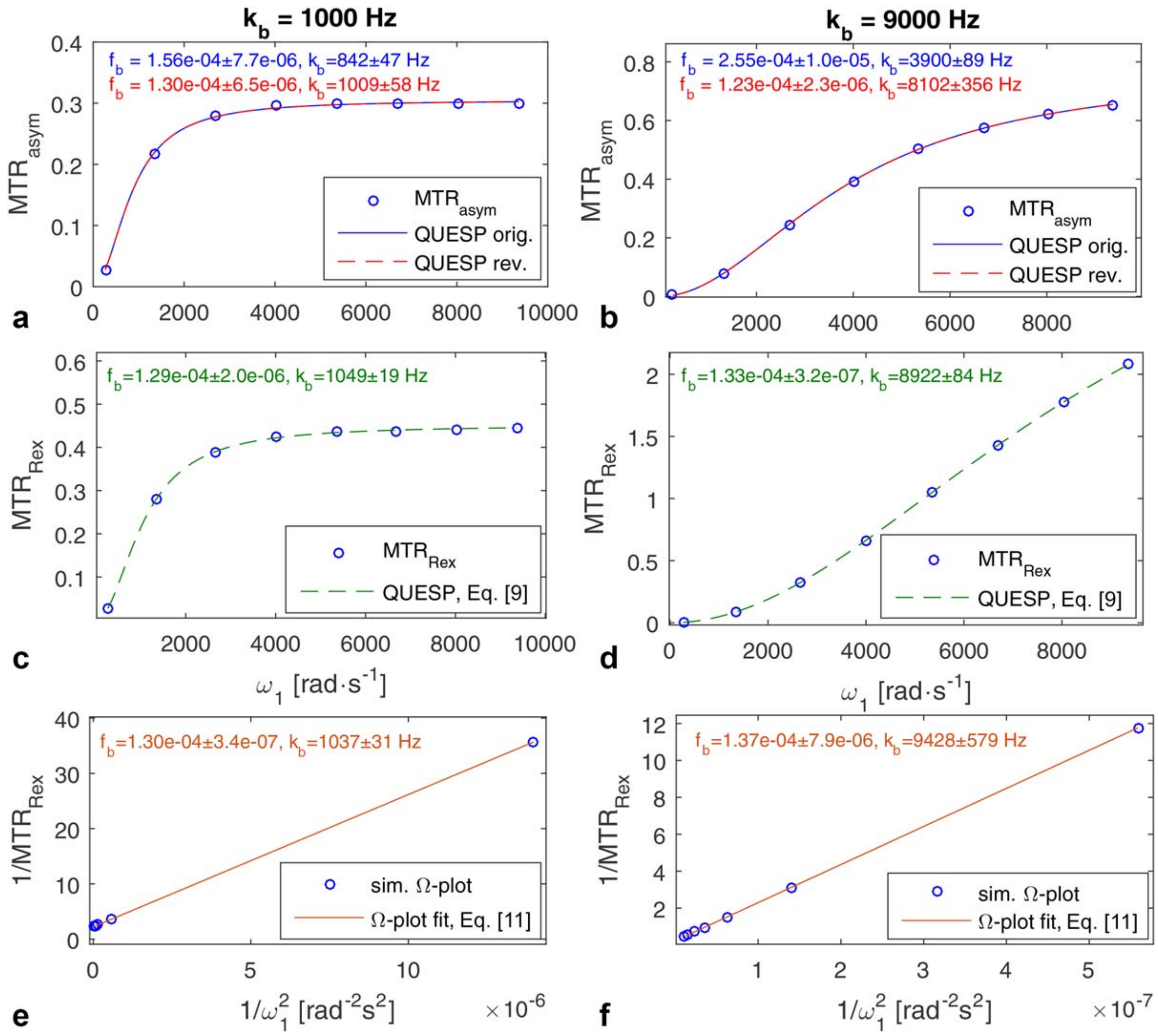

FIG. 3. Different QUESP methods applied to simulated data for $f_b = 1.35 \times 10^{-4}$, $k_b = 1000$ Hz (left), and $k_b = 9000$ Hz (right). **a, b**: The original QUESP equation in steady-state (blue solid line, Eq. [6]) and the revised QUESP equation in steady-state (red dashed line, Eq. [4]). Although both equations fit the data, the revised equation yields much better estimates for exchange rates, especially for lower labeling (faster exchange). **c, d**: The QUESP method in steady state using the inverse metric approach (green dashed line, Eq. [9]). **e, f**: Ω-plot method in steady state (Eq. [10]).

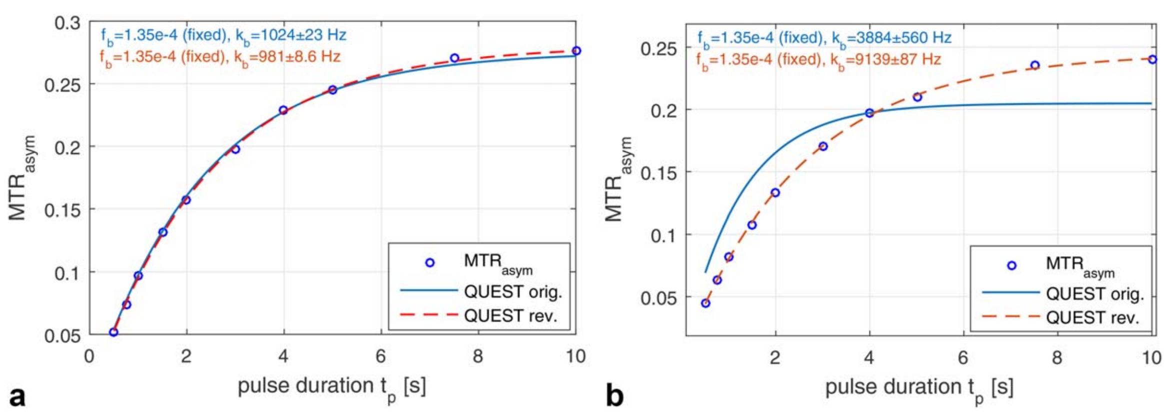

FIG. 4. The QUEST fitting with original and revised equation (Eqs. [6] and [4], respectively) for $k_b = 1000$ Hz (**a**) and $k_b = 9000$ Hz (**b**). Estimation is improved by the revised formula especially for higher exchange rates. Note: When a starting value of $k_b = 4000$ Hz was chosen, the fit for MTR ($k_b = 1000$ Hz) yielded a completely wrong estimation, because of the two possible values of k_b with the same MTR (see Fig. 2f).

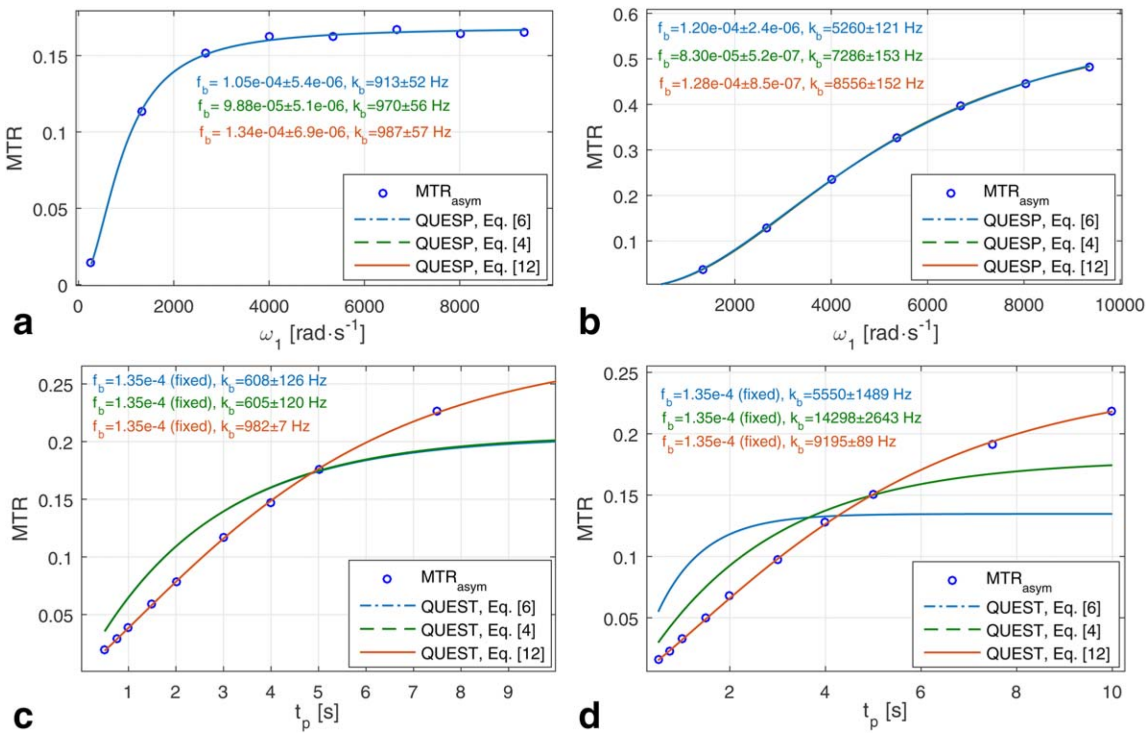

FIG. 5. **a, b**: Different QUESP evaluations applied to nonequilibrium simulated data for defined $Z_i = 0.3$, short $t_p = 3$ s, relative concentration $f_b = 1.35 \times 10^{-4}$, and exchange rates $k_b = 1000$ Hz (**a**) and $k_b = 9000$ Hz (**b**). Generally, all of the formulas fit the QUESP data; however, both Equations [6] and [4] lead to wrong parameter estimations. The fit of the data leads to meaningful estimations only if the initial magnetization is properly incorporated in the model (Eq. [12]). **c, d**: Different QUEST evaluations applied to nonequilibrium simulated data for defined $Z_i = 0.3$, relative concentration $f_b = 1.35 \times 10^{-4}$, and exchange rates $k_b = 1000$ Hz (**c**) and $k_b = 9000$ Hz (**d**). Equation [12] (red line) is able to fit the data with good exchange rate estimation; again, the models that do not incorporate the initial magnetization fail in predicting the correct exchange rates.

needed to be fulfilled to get accurate parameter estimation. To be able to interpret speed-up experiments, QUESP and QUEST estimations with arbitrary initial magnetization before saturation were also considered. We assumed the same system as before (Figs. 3a and 3b), yet using an initial magnetization of $M_i = 0.3 \times M_0$ or $Z_i = 0.3$ to perform QUESP (Figs. 5a and 5b). Using the revised equation including Z_i (Eq. [12]), the estimation and fitting procedure improves significantly, resulting in values close to the initially provided ones. It is important to note that different QUESP equations actually fit the data very well. However, the estimated parameters are very different across different equations used. In turn, the plausibility of obtained results is seriously compromised, despite the "good" fits to the given data.

When the QUEST estimations are performed for Zi $\neq$ 0, the outcome is similar for QUESP at $M_i = 0.3 \times M_0$ or $Z_i = 0.3$ (Figs. 5c and 5d). If M_i and M_0 are provided, the extended Eq. [12] yields good estimates. In contrast, all other equations yield worse fits and biased estimations for exchange rates.

Experimental Results

Following the calculations with simulated parameters, we performed a series of multi-B_1 experiments with EuDOTAM-Gly between 10 and 30°C with the saturation having reached a steady state (Figs. 6 and 7). The three-pool BM fits show a good match to the Z-spectra, and exchange rates are determined to be approximately 3000 Hz at 13.4°C and approximately 7000 Hz at 23.8°C (Figs. 6a and 6b). When the original QUESP formula was used, an exchange rate of 2000 and 3200 Hz at 13.4 and 23.8°C, respectively, were obtained. Finally, the improved formula for QUESP and the Ω-plot method yielded much better exchange-rate estimations of 2700 and 6000 Hz at 13.4 and 23.8°C, respectively. When the estimations are expanded to four sets of experimental data with temperatures ranging between 10 and 30°C, and results are compared across the different methods (Fig. 7a), the exchange process (evaluated by the Arrhenius parameters, Eq. [13] can be determined with best accuracy using the BM fit ($E = 59.4 \pm 10$ kJ/mol, $k_c = 7657 \pm 495$ Hz), followed by the revised Ω-plot and QUESP methods. The original QUESP equation (Eq. [6]) yields strong underestimations of the Arrhenius exchange parameters $E = 30.7 \pm 8.6$ kJ/mol, $k_c = 3379 \pm 168$ Hz (data not shown).

If an even larger temperature range is used (T ≈ 10–50°C) (Fig. 7b), the exchange rates increase up to 50 kHz. This violates the required condition "chemical shift difference > CEST peak width" (31), and thus the regime of validity of the simplified formulas for the labeling efficiency (Eqs. [A7] and [A5]), as discussed subsequently. In such case, only the full BM equation yields reliable results and matches well with the Arrhenius Equation

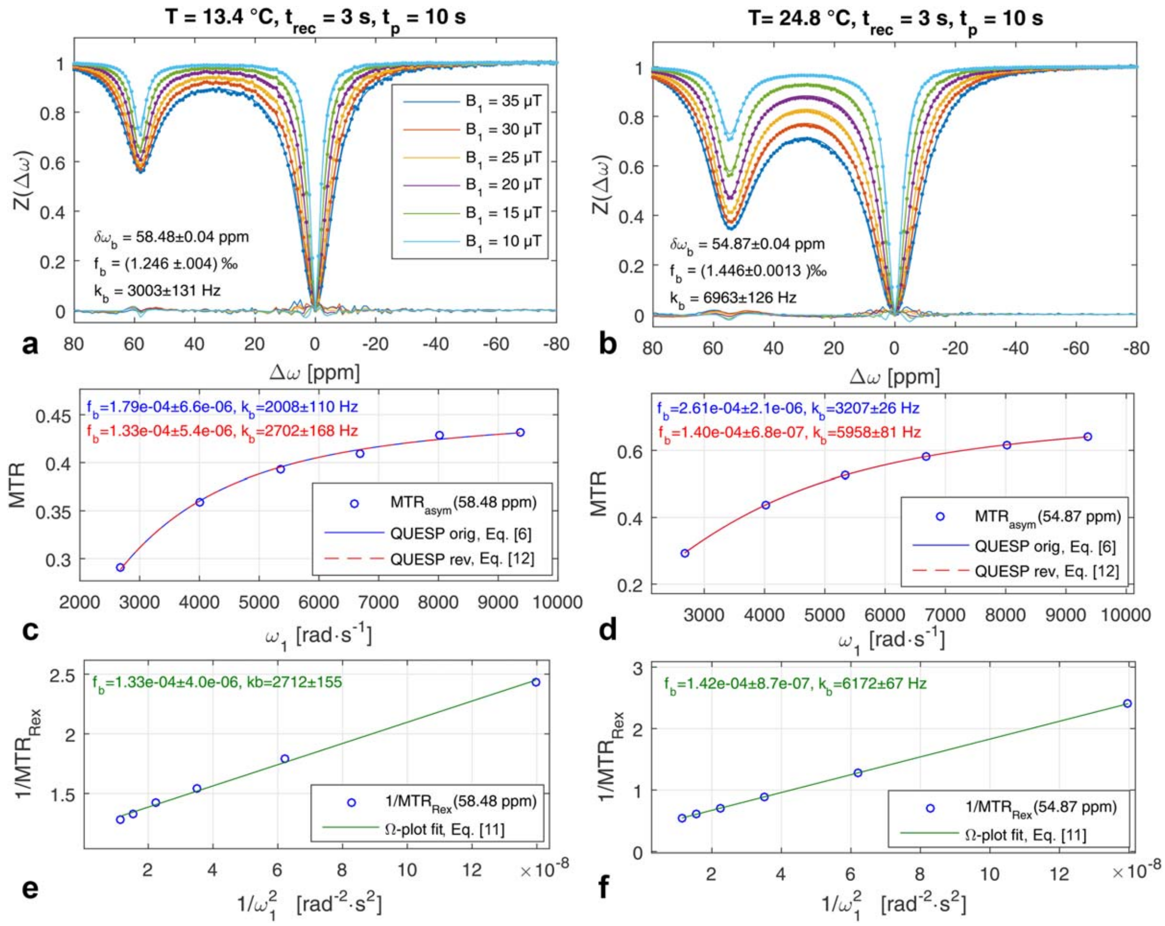

FIG. 6. Different steady-state QUESP methods applied to experimental data T = 13.4°C (left) and T = 23.8°C (right). **a, b**: Multi-B_1 Z-spectral data (dots) fitted with the BM equation (solid lines), showing good match with small residuals (dotted lines at Z ≈ 0). **c, d**: Fitting results according to the original QUESP Equation [6], showing underestimation of the exchange rates; however, better estimations are achieved using the revised QUESP Equation [4] or the Ω-plot method (**e, f**).

[13] at all temperatures, followed in accuracy by the Ω-plot and finally the revised QUESP method (Fig. 7b).

To show the improvement in the regime of short saturation and arbitrary initial conditions, Z-spectra with a recovery time of only 1 s and a saturation time of 3 s were acquired for the temperatures 13.4 and 23.8°C (Fig. 8). Because of the short recovery, the initial magnetization before saturation was measured to be $Z_i = 0.865$ at 13.4°C and $Z_i = 0.769$ at 23.8°C. This was provided to the BM fit algorithm along with the measured R_{1a} values

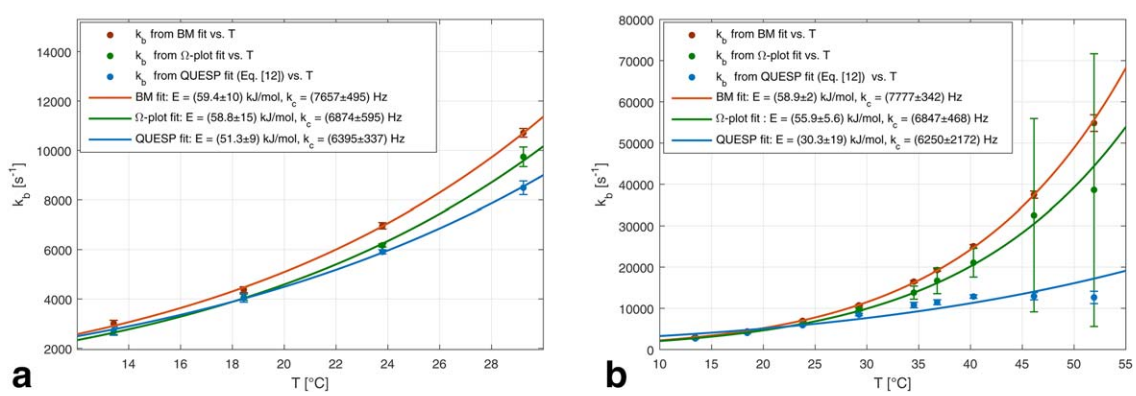

FIG. 7. Comparison of steady-state QUESP and Ω-plot methods with full BM fit results (data points) together with corresponding temperature fits obtained using the Arrhenius equation (Eq. [13]). **a**: Temperature range between 12 and 30°C; Both the Ω-plot and revised QUESP methods yield reasonable results close to the result obtained from the BM fit. **b**: Temperature range from 12 to 50°C: Only the BM fit shows good accuracy, whereas the QUESP method deviates already above 10^4 Hz and the Ω-plot method deviates above $2{\cdot}10^4$ Hz. The k_c values obtained according to the BM equation match well with the Arrhenius fits for both temperature ranges.

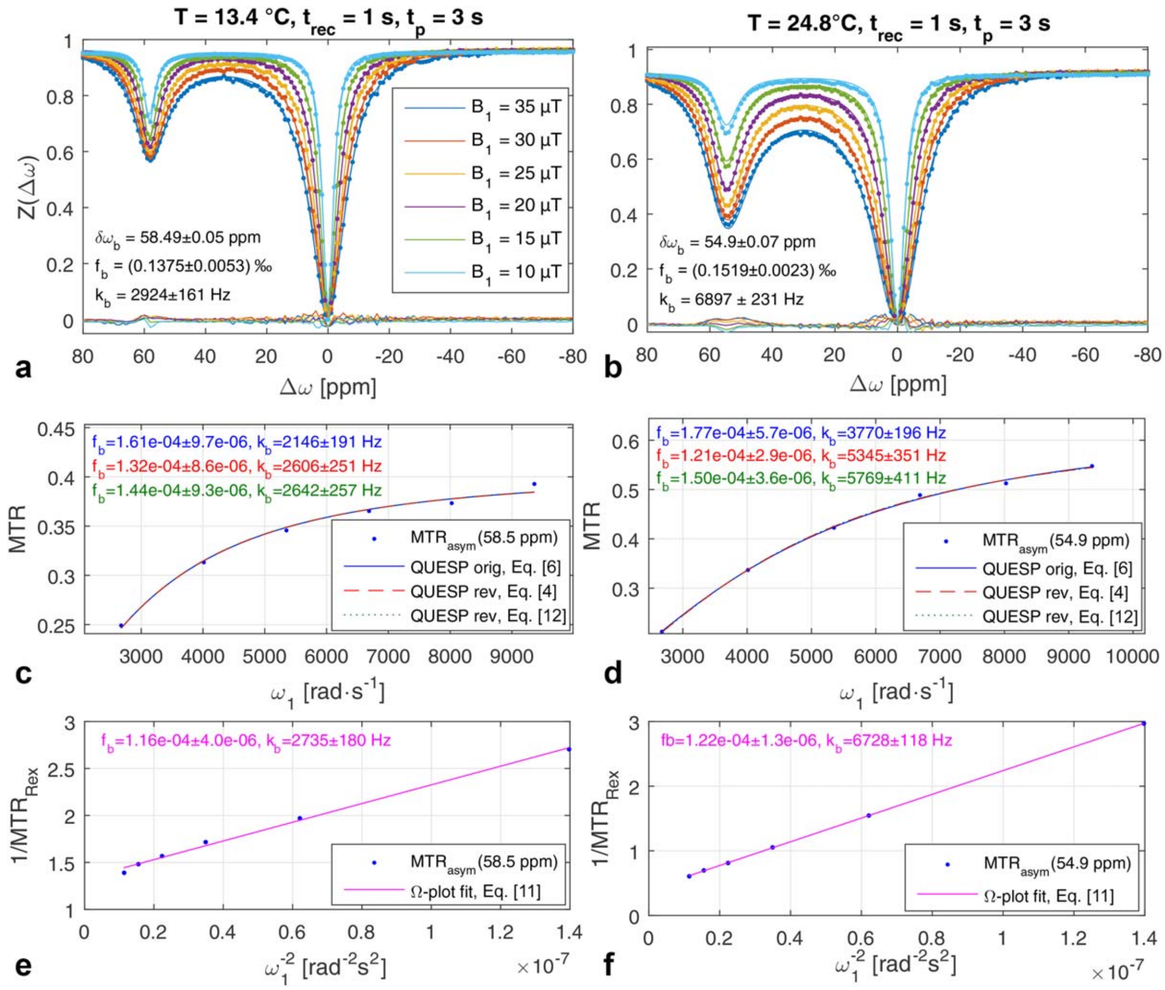

FIG. 8. Different QUESP methods applied to experimental data T = 13.4°C (left) and T = 23.8°C (right) for initial conditions $Z_i < 1.0 = 0.87$ (**a**) and $Z_i = 0.77$ (**b**). Multi-B_1 Z-spectral data (dots) fitted with the BM equation (solid lines) show good match with small residuals (dotted lines at $Z \approx 0$). **c, d**: Fitting results according to the original QUESP equation show underestimation of the exchange rates, whereas better estimations are achieved using the revised QUESP, including Z_i (Eq. [12]) or the Ω-plot method (**e, f**).

of 0.42 and 0.33 Hz, respectively, to allow fitting of the multi-B_1 Z-spectra, showing a baseline of less than 1 (Figs. 8a and 8b). The results of this speed-up experiment match well with the steady-state outcome (Fig. 6). For the evaluation of the improved theory, we compared the original equation with the revised equation for $Z_i = 1$ (Eq. [4]), and the revised equation for arbitrary Z_i (Eq. [12]). For both temperatures, Equation [12] shows the best results when compared with the full BM fit (Figs. 8c and 8d). For the lower temperature, the improvement is negligible, as Z_i is already close to 1 because of the faster T_1 relaxation, whereas the improvement becomes significant for 23.8°C. Interestingly, we note that the Ω-plot method still yields relatively good exchange-rate estimation, despite being used outside of its expected convergence regime of steady-state saturation (Figs. 8e and 8f).

DISCUSSION

This work is an extension of the quantitative analytical description of the QUEST and QUESP methods (21,23,38), improving the case of weak labeling ($\gamma B_1 < k$). These results show that this is especially important when evaluating faster exchange rates, which often result in lower labeling efficiencies. In addition, we extended the formulas for arbitrary initial conditions for the Z-magnetization. Before that, the design of a quantitative CEST experiment required a fully relaxed system (i.e., a system that has reached saturation) or a system that is close to a saturation steady state, to simplify the data analysis. However, these experimental settings are time-consuming and the results are biased in the case of non-equilibrium initial magnetization. In this work, we demonstrated that the analysis of data with arbitrary initial condition yields accurate estimates of exchange rates only when the initial magnetization is known and taken into account. This is not only important when analytical evaluation formulas are used, but also when full numerical BM equations are used; the insight of the analytical formulas shows that the initial magnetization needs to be known. As shown in the Supporting Information, different normalization can hide this necessity, leading to fits matching the data, but giving poor estimations. Furthermore, considering the existence of several evaluation methods, we compared the methods that have been widely used in the literature to assess which method is the best (i.e., the most convenient), without compromising the accuracy of the obtained results.

To this end, if scanning time is not an issue, the best quantification is clearly achieved with multiple B_1–Z-spectra fitting using the BM equation. Alternatively, if the scan time is limited and only a certain number of offsets can be acquired, we show here that the revised QUESP and Ω-plot methods yield estimations close to the real value in simulations or the BM fit results in experiments. Compared with fitting single offset QUESP data using the analytical expression, the Ω-plot method additionally cancels spillover effects (33), as it is implicitly based on the spillover-corrected inverse metric of the steady-state Z-spectrum (27,39). Although this is a steady-state method requiring long saturation, its features and the resulting accuracy shown here suggest the Ω-plot method as the method of choice for many quantitative CEST applications. A limitation of this method is the number of pools that can be accurately quantified (if handling two or more pools with similar frequencies, the linearity of the Ω-plot is lost and simple linear regression is not possible anymore) (see Eq. [10]). In such case, a linear superposition of QUESP (Eq. [4]) might be more promising. We want to point out that Sun et al. showed that the spillover-correcting inverse metric approach is also applicable in non–steady state, as the transient state of the inverse metric decays relatively quickly (40). This could explain why we still observed good estimation of exchange rates in the speed-up experiment (Fig. 8).

In case the scan time needs to be reduced even more, the revised QUESP method for arbitrary conditions (Eq. [12]) allows for reliable interpretation of data. Here, the prerequisite is knowledge of the initial magnetization M_i just before the saturation; then the saturation does not need to reach steady state and can still be evaluated. Consequently, we demonstrated that, in addition to the full BM equation, only the revised QUESP equation can lead to correct results, when the M_i is provided (Fig. 4). We point out that similar findings on the importance of initial condition were also previously reported by Yuwen et al. (41).

It is often stated in the community that both QUESP and QUEST are fully quantitative methods. However, as seen in Equations [A3] and [A10], QUEST can actually determine only R_{ex}, whereas QUESP can alter R_{ex} and thus separate k_b and f_b. Consequently, QUEST experiments require either the concentration of exchanging species as an input, or the need to scan the whole frequency range and use implicitly the peak width as a second independent access to k_b. In fact single resonant QUEST experiment can never separate k_b and f_b, if not also acquired at different saturation powers. Because the separation of k_b and f_b is only possible by QUESP, we recommend that quantitative CEST experiments are performed using variable saturation power. Important to note is that QUESP requires the longitudinal relaxation rate of water R_{1a} for absolute concentration determination. Additionally, for single exchanging pools, on-resonant spin-lock also allows for accurate quantification of exchange rates and can provide an alternative measurement to CEST (25).

ParaCEST Results

The estimated exchange rates for EuDOTAM-Gly (Fig. 6) are in line with the previously published results. We obtained k_b between 6000 and 7600 Hz at 25°C with the different methods, which matches the originally reported value of 6250 Hz at 25°C (27). Interestingly, the concentrations obtained from the fits were slightly higher ($f_b = 1.2$–$1.4‰$) than actually used in experiments ($f_b = 0.9‰$); however, results were consistent for the BM fit as well as the QUESP methods. For higher temperature, the breakdown of the QUESP methods could be assessed using the underlying Equation [A7] of α, which is only valid in the narrow-peak limit $\delta\omega >> \Gamma$, in which $\delta\omega$ is the chemical shift difference and Γ is the B_1-dependent width of the CEST peak (Eq. [A6] that depends on the exchange rate (31). For the paraCEST agent used in this study, we conclude that the exchange rates should be smaller than 10000 Hz for the QUESP/QUEST and Ω-plot methods to be valid at a field strength of 7 T (i.e., the agent is resonating at 55 ppm at 7 T, which gives $\delta\omega = 7\cdot\gamma\cdot 50\,\text{ppm} \approx 10^5\ s^{-1}$ and $\Gamma(k = 10^4\,Hz, B_1 = 25\,\mu T) \approx 2 \times 10^4\,s^{-1}$).

In principle, the QUESP equation can be extended by the full term for labeling efficiency, resulting in Equation [14], as follows (31):

$$\alpha = \frac{\frac{\delta\omega_b^2}{\omega_1^2+\delta\omega_b^2}k_b + R_{2b}}{k_b + R_{2b}} \frac{\omega_1^2}{\omega_1^2 + k_b(k_b + R_{2b})} \qquad [14]$$

This should potentially extend the convergence interval of the QUESP method for higher exchange rates or smaller chemical shifts, such as salicylic compounds (42,43) or endogenous metabolites (31,44). We want to point out that we neglected R_{1b} and R_{2b} in all evaluations with respect to the fast exchange rates; however, especially for lower temperatures, CEST pool relaxation rates might not be negligible for fitting paraCEST spectra. Thus, although R_{2b} is already included (26), the analytical description of the labeling efficiency still needs to be extended by R_{1b}, improving. The original definition of α (in (23), $\alpha = \omega_1^2/(\omega_1^2 + pq)$) is not valid in the case of the exchange rates of paraCEST agents ($k_b > R_{1a}/f_b$; for more details see Supporting Information).

Optimal B_1 and t_p Ranges

When the experiments are set up, the most common question is which B_1 values and saturation times should be used for optimal sampling of the QUESP and QUEST experiments. Evaluation of QUESP becomes easier in steady state and long t_p, but signals can be higher for lower saturation times, especially in the presence of spillover (45). To improve separation of k_b and f_b, the B_1 values in QUESP need to sample the varying labeling efficiency α (Eq. [A7]) as well as possible. This is the case if B_1 are sampled around $B_1 = k/\gamma$ (e.g., $0.25\cdot k/\gamma$, $0.5\cdot k/\gamma$, $1\cdot k/\gamma$, $1.5\cdot k/\gamma$, and $2\cdot k/\gamma$ would lead to $\alpha = 6$, 20, 50, 69, and 80%). Of course, this requires an estimate of the exchange rate, but it is plausible that sampling around 50% ($B_1 = 1\cdot k/\gamma$) of the maximal effect is close to optimal. In principle, two B_1 values are enough to solve the equations, but more sampling improves the estimation. As shown in Figure 2b, using too low B_1 powers hinders the determination of fast exchange rates, and using too high B_1 powers hinders the determination of slower exchange rates. The same trend is shown in Figure 2f, where MTR shows a plateau, depending on B_1, meaning particular MTR(B_1,k_b) regions are almost independent of k_b.

The same curve in Figure 2f gives insight into the QUEST measurement setup: B_1 values should also be chosen so that the expected exchange rate does not fall into the plateau. It is important to note that two solutions for k_b can exist for the same QUEST course, one being left from the optimum and one right from the optimum (Fig. 2f). This makes the correct choice of B_1 for QUEST even more important. Finally, the saturation time t_p should sample the saturation build-up, which is typically below 1 to 2 × T_1. Beyond these rules of thumb, the optimal sensitivity of the CEST sequence can be derived by combining the analytical formula with the approach of Jiang et al. for steady-state sequences (46).

Radiation Damping

We want to point out that data acquisition (especially FID at high magnetic fields) for nonequilibrium initial condition or non–steady state (so especially QUEST) are prone to errors originating from altered T_1 relaxation during recover and saturation time as a result of radiation damping effects (47). As radiation damping and thus the effective T_1 depends on the size of the magnetization vector, the QUEST method can be drastically affected by this effect, and estimated exchange rates are questionable (see also Supporting Fig. S4). We were able to reduce this effect by an order of magnitude using a smaller NMR tube for the shown accelerated sequence (Fig. 8).

2π or Not 2π

When a QUESP data set or an Ω-plot is evaluated, the exchange rates are obtained in units of ω_1, which is rad·s^{-1}. Therefore, the question arises whether the final results of k_b have to be converted from rad·s^{-1} to s^{-1}, by dividing by 2π. By looking into the fundamental BM Equation [A6], one notes that ω_1 and T_1, T_2, or the exchange rate k_b appear within the same equation, although with different units of rad·s^{-1} and s^{-1}, respectively. Thus, the value of oscillation constants corresponds directly to the value of the exchange rate in s^{-1}, and we conclude that no further division by 2π should be performed to obtain the final value of the exchange rate.

Spillover Effect, Multiple Pools, and Pulsed Saturation

The quantification methods in this work were only applied in a paraCEST system in which direct water saturation is not an issue. Albeit not shown here directly, all equations based on the inverse metric (i.e., MTR_{Rex} in Eqs. [3] and [8]) will still be valid, as long as saturation steady state is reached (33,39,48). In contrast, all dynamic solutions will not be directly valid in the case of strongly overlapping direct water saturation. In such a case, our methodology can be extended using the QUESTRA approach of Sun et al. (49).

Moreover, pulsed saturation can be incorporated for steady-state conditions as shown previously (32,33). However, when a single offset approach is being used, the interplay of several CEST resonances further confounds the obtained results. Finally, R_{2b} values should generally be taken into account for all of the presented approaches, which is easily possible; however, in the case of paraCEST agents, these contributions are negligible.

CONCLUSIONS

How do we accurately measure exchange rates using CEST? In this article, we attempted to address the CEST quantification issue by providing revised equations to be used in respective calculations, discussing different theoretical and experimental cases, and finally providing some practical hints. In addition to improving the theory by adding missing labeling factors, we were able to accurately estimate exchange rates and extend the analytic models for the case of arbitrary initial magnetization and, subsequently, non-steady-state saturation. By completing the quantitative description, we could show a methodology that allows for more freedom in choosing the appropriate quantitative CEST experiments, and speed up the quantitative experiments without biasing the final results.

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APPENDIX

The CEST results are presented in so-called Z-spectra. To obtain a Z-spectrum, the amplitude of the saturated water peak ($M_{sat}(\Delta\omega)$), normalized as a proportion of the water signal in thermal equlibrium (M_0), is plotted as a function of the saturation pulse frequency offset from water Δω. The normalized Z-spectrum is described by the function Z(Δω) given by

$$Z(\Delta\omega) = M_{sat}(\Delta\omega)/M_0 \quad [A1]$$

In the case of continuous saturation with a block pulse of amplitude $B_1 = \omega_1/\gamma$ and duration t_p, the normalized water Z-magnetization Z(Δω) has been derived previously (31) and is given with Equation [A2], in which Z_i is the initial Z-magnetization before saturation and can be written as $Z_i = M_i/M_0$.

$$Z(\Delta\omega, t_p) = Z^{ss}(\Delta\omega) + (\cos^2\theta \cdot Z_i - Z^{ss}(\Delta\omega))e^{-R_{1\rho}(\Delta\omega)t_p} \quad [A2]$$

The measured $Z(\Delta\omega, t_p)$ decays or grows during the saturation module toward a steady state, given by

$$Z^{ss}(\Delta\omega) = \cos^2\theta \frac{R_{1a}}{R_{1\rho}(\Delta\omega)} \quad [A3]$$

where the longitudinal relaxation rate of water is R_{1a}; the longitudinal relaxation rate in the rotating frame of water is $R_{1\rho}$; and the tilt angle of the effective field is $\theta = \tan^{-1}(\omega_1/\Delta\omega)$. In the large-shift limit (chemical shift of the CEST pool $\delta\omega_b >> \omega_1$, k_b) and without semisolid MT, some terms simplify to $\cos^2\theta=1$, $R_{1\rho}=R_{1a}+R_{ex}(\Delta\omega_b)$ and close to the CEST resonance, so the Z-value can be defined as

$$Z(\Delta\omega_b, t_p) = \frac{R_{1a}}{R_{1a}+R_{ex}(\Delta\omega_b)} + \left(Z_i - \frac{R_{1a}}{R_{1a}+R_{ex}(\Delta\omega_b)}\right)e^{-(R_{1a}+R_{ex}(\Delta\omega_b))t_p} \qquad [A4]$$

where $R_{ex}(\Delta\omega_b)$ is the exchange-dependent relaxation that simplifies to Equation [A5] for narrow peaks ($\delta\omega_b > \Gamma \approx 2\sqrt{\omega_1^2+k_b{}^2}$) to a Lorentzian line centered at the frequency offset relative to the CEST pool $\Delta\omega_b$, with the linewidth Γ defined by Equation [A6] and the value of the label scan, $R_{ex}^{lab} = R_{ex}(\Delta\omega = \delta\omega_b) = R_{ex}(\Delta\omega_b = 0)$, defined by Equation [A7].

$$R_{ex}(\Delta\omega_b) = \frac{R_{ex,LS}^{lab}\,{}^{\Gamma^2}/_4}{{}^{\Gamma^2}/_4 + \Delta\omega_b^2} \qquad [A5]$$

$$\Gamma = 2\sqrt{\frac{R_{2b}+k_b}{k_b}\omega_1^2 + (R_{2b}+k_b)^2} \qquad [A6]$$

$$R_{ex}^{lab} = f_b k_b \cdot \alpha = f_b k_b \frac{\omega_1^2}{\omega_1^2 + k_b(k_b+R_{2b})} \qquad [A7]$$

The reference value at the opposite frequency can often be assumed as zero according to Equation [A8], if $\Delta\omega >> \Gamma$:

$$R_{ex}^{ref} = R_{ex}(\Delta\omega = -\delta\omega_b) = R_{ex}(\Delta\omega_b = -2\delta\omega_b) \approx 0 \qquad [A8]$$

The exchange-dependent relaxation R_{ex} is the most important parameter for CEST effects and can be used to determine the exchange rate k_b of the exchangeable protons with concentration fraction f_b and transverse relaxation R_{2b}. Please note that, compared with R_{ex} defined by Trott and Palmer (50) or Jin et al. (25), the R_{ex}^{lab} term here includes a $\sin^2\theta$ term, so $R_{ex}^{lab} = \sin^2\theta \cdot R_{ex}$. This is necessary to derive the known labeling efficiency α.

The magnetization transfer ratio asymmetry (MTR_{asym}) is defined as the difference in signal on either side of the water peak centered at 0 ppm for constant amplitude B_1, according to Equation [A9], in which the label and reference normalized Z-magnetizations are $Z_{lab} = Z(\Delta\omega = +\delta\omega_b)$ and $Z_{ref} = Z(\Delta\omega = -\delta\omega_b)$, respectively.

$$MTR_{asym} = Z_{ref}(t_p, B_1) - Z_{lab}(t_p, B_1) \qquad [A9]$$

Using Equation [A4], the MTR_{asym} yields Equation [A10] as follows:

$$MTR_{asym} = \frac{R_{ex}^{lab}}{R_{1a}+R_{ex}^{lab}} + (Z_i - 1)e^{-R_{1a}t_p} - \left(Z_i - \frac{R_{1a}}{R_{1a}+R_{ex}^{lab}}\right)e^{-(R_{1a}+R_{ex}^{lab})t_p} \qquad [A10]$$

This asymmetry analysis provides a clear representation of the transient CEST effect, assuming that direct water saturation is not overlapping strongly with the CEST peak.

Equations [A4] and [A7] show that two experiments can be designed to quantify exchange rates independently from the water signal, namely quantification of exchange by varying saturation time t_p (QUEST) and saturation power B_1 (QUESP).

Equation [A9] in steady state ($t_p >> R_1$) is further simplified to

$$MTR_{asym} = Z_{ref}(B_1) - Z_{lab}(B_1) = \frac{f_b k_b \cdot \alpha}{R_{1a} + f_b k_b \cdot \alpha} \qquad [A11]$$

In the case of overlapping direct water saturation, MTR_{Rex} is a useful metric that was derived previously for steady-state conditions (26) and can be written as

$$MTR_{Rex} = \frac{1}{Z_{lab}(B_1)} - \frac{1}{Z_{ref}(B_1)} = \frac{f_b k_b \cdot \alpha}{R_{1a}} \qquad [A12]$$

Some in vivo applications of CEST are shown to benefit from the fact that MTR_{Rex} removes T_2 and magnetization transfer dependencies, and hence directly yields measurements of R_{ex} (39). In addition, the plot of $1/R_{ex}$ as a function of $(1/\omega_1)^2$ yields a linear function according to Equation [A7] which can be used as another approach for exchange-rate quantification and is known as the Ω-plot method (see also Eqs. [10] and [11].

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SUPPORTING INFORMATION

Additional Supporting Information can be found in the online version of this article.

Fig. S1. The MTR_{asym} of simulated data (circles) together with analytic solutions of Equation [2] (solid lines) in the case of normalization by M_0 (**a–c**) and normalization by $M_{offres} = 0.71 \times M_0$ (**d–f**). **g–i**: Both the data and the analytical formula (dashed line, Eq. [3]) are normalized by M_{offres}.

Fig. S2. The MTR_{asym} of simulated data normalized by individual $M_{offres}(t_p)$ (circles) together with differently normalized analytic solutions (lines). **a–c**: Solid lines show Equation [2] with M_0 normalization. **d–f**: Dashed lines show Equation [3] with $M_{offres}(3s) = 0.71 \cdot M_0$ normalization. **g–i**: Dash-dotted lines show Equation [4] with individual $M_{offres}(t_p)$ normalization.

FIG. S3. **a–c**: The MTR_{asym} of simulated data normalized by individual $M_{offres}(t_p)$ (circles) with the original Equation [1] without Z_i. **d–f**: Equation [1] was normalized manually by a factor 1.19 to show that the analytic solution (Eq. [1]) yields a relatively good match to the data.

FIG. S4. T_1 inversion recovery data without (red lines) and with (blue lines) gradient applied during the inversion time. By applying the gradient radiation, damping is avoided. Comparison of (**a**) and (**b**) reveals that going from a 5-mm tube to a 1.6-mm tube solves this issue.